

EXP NO : 7

DATE : 23/03/2026

BUILDING AGGREGATION PIPELINES USING MONGODB COMPASS

```
>_MONGOSH
> use companyDB
< switched to db companyDB
> db.createCollection("employees")
< { ok: 1 }
> db.employees.insertMany([
  { name: "Alice", dept: "HR", salary: 30000, age: 25 },
  { name: "Bob", dept: "IT", salary: 40000, age: 30 },
  { name: "Charlie", dept: "Finance", salary: 35000, age: 28 },
  { name: "David", dept: "IT", salary: 45000, age: 32 },
  { name: "Eve", dept: "HR", salary: 32000, age: 26 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f6125aeed29aa093adb722'),
    '1': ObjectId('69f6125aeed29aa093adb723'),
    '2': ObjectId('69f6125aeed29aa093adb724'),
    '3': ObjectId('69f6125aeed29aa093adb725'),
    '4': ObjectId('69f6125aeed29aa093adb726')
  }
}
> db.employees.aggregate([
  { $match: { dept: "IT" } }
])
< {
  _id: ObjectId('69f6101dc15240977f29ba7a'),
  emp_id: 2,
  name: 'Bob',
  dept: 'IT',
  salary: 40000,
  city: 'Delhi'
}
{
  _id: ObjectId('69f6101dc15240977f29ba7c'),
  emp_id: 4,
  name: 'David',
  dept: 'IT',
  salary: 45000,
```

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>_MONGOSH
{
  _id: ObjectId('69f6125aeed29aa093adb723'),
  name: 'Bob',
  dept: 'IT',
  salary: 40000,
  age: 30
}
{
  _id: ObjectId('69f6125aeed29aa093adb725'),
  name: 'David',
  dept: 'IT',
  salary: 45000,
  age: 32
}
> db.employees.aggregate([
  { $group: { _id: "$dept", totalSalary: { $sum: "$salary" } } }
])
< {
  _id: 'Finance',
  totalSalary: 70000
}
{
  _id: 'HR',
  totalSalary: 124000
}
{
  _id: 'IT',
  totalSalary: 170000
}
> db.employees.aggregate([
  { $group: { _id: "$dept", avgSalary: { $avg: "$salary" } } }
])
< {
  _id: 'Finance',
  avgSalary: 35000
}
{
  _id: 'HR',
```

```

>_MONGOSH
{
  _id: 'IT',
  avgSalary: 42500
}
> db.employees.aggregate([
  { $project: { name: 1, salary: 1, _id: 0 } }
])
< {
  name: 'Alice'
}
{
  name: 'Charlie'
}
{
  name: 'Alice'
}
{
  name: 'Charlie'
}
{
  name: 'Alice',
  salary: 30000
}
{
  name: 'Bob',
  salary: 40000
}
{
  name: 'Charlie',
  salary: 35000
}
{
  name: 'David',
  salary: 45000
}
{
  name: 'Eve',
  salary: 32000
}

```

```

> db.employees.aggregate([
  { $match: { salary: { $gt: 30000 } } },
  { $group: { _id: "$dept", totalSalary: { $sum: "$salary" } } },
  { $sort: { totalSalary: -1 } }
])
{
  _id: 'IT',
  totalSalary: 170000
}
{
  _id: 'Finance',
  totalSalary: 70000
}
{
  _id: 'HR',
  totalSalary: 64000
}
companyDB > |

```